Expert Systems With Applications 318 (2026) 132001

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Dynamic cognitive cycle-driven multimodal agent for knowledge graph completion

Kai Yang [a], Xinzhi Wang [a], Baoguo Lu [b], Xiangfeng Luo [a,*], Chengfan Li [a], Jianqiang Huang [a]

[a] Knowledge Science and Intelligent Generation Laboratory, Shanghai University, No. 99 Shangda Road, Shanghai, 200444, Shanghai, China
[b] Nanjing Research Institute of Electronics Engineering&Nanjing University, 163 Xianlin Avenue, Nanjing, 210023, Jiangsu, China

ARTICLE INFO

Keywords:
Multimodal knowledge graph completion
Intelligent agents
Perception
Memory mechanisms
Self-alignment learning

ABSTRACT

Multimodal Knowledge Graph Completion (MMKGC) is essential for addressing information scarcity in structured knowledge bases. However, conventional methods typically adhere to a static embedding paradigm, where fixed entity representations fail to capture query-specific semantics, limiting reasoning flexibility. To overcome this, we propose a context-aware MMKGC Agent built on a tightly coupled, closed-loop cognitive architecture. Unlike modular pipelines that rely on the sequential stacking of encoders, our agent establishes a functional interdependency between representational plasticity and semantic stability. Specifically, the agent initiates a dual-path perception to extract global-local structural cues, which are subsequently transformed into dynamic memory states via a Learnable Residual Memory Pathway (LRMP). This allows the agent to actively synthesize internal representations tailored to the specific intent of each query. To counterbalance the representational drift inherent in such dynamic adaptation, we integrate a Self-Alignment Stabilization system that utilizes a redundancy-reducing objective to serve as a semantic anchor. This mechanism ensures that the agent's adaptive reasoning remains intrinsically coherent and robust against multimodal noise. Extensive experiments on four benchmarks (DB15K, MKG-W/Y, KVC16K) demonstrate that this synergistic paradigm consistently surpasses state-of-the-art baselines, particularly in resolving complex, ambiguous reasoning tasks where static models fail.

1. Introduction

Multimodal Knowledge Graphs (MMKGs) (Chen et al., 2024; Gong et al., 2024) provide a unified representation of the world by combining the structured logic of triples with the perceptual richness of multimodal data such as images and textual descriptions. This integration enables MMKGs to capture comprehensive semantics and support advanced applications in recommendation systems (Sun et al., 2020), computer vision, and natural language processing (Chen et al., 2023). Despite this potential, MMKGs are inherently incomplete due to the scarcity of multimodal corpora and the complexity of modality alignment, leading to missing structural and semantic information. Consequently, MMKGC (Chen et al., 2024) has emerged as a critical research focus.

Existing MMKGC methods (Cao et al., 2022b; Lee et al., 2023; Li et al., 2023; Moussely-Sergieh et al., 2018a; Zhang et al., 2025), however, largely follow a **static embedding paradigm**, in which each entity is assigned a single, context-independent representation after initial encoding. Specifically, entity embeddings are typically produced by pretrained multimodal encoders (e.g., BERT (Devlin et al., 2019) for textual descriptions or ImageNet-pretrained CNNs for images) whose outputs are often kept frozen and reused across different queries; hence the representation of an entity h in $(h, r_1, ?)$ is identical to that in $(h, r_2, ?)$. Although these methods incorporate multimodal signals, the fusion is often coarse-grained (e.g., instance-level averaging or modality-level pooling) and heavily relies on fixed pretrained features, limiting their capacity to model fine-grained, query-sensitive entity–relation interactions. Consequently, such static representations struggle to adaptively align with the specific intent of varying queries, constraining the reasoning capability in complex multimodal environments.

To address these challenges, we propose a context-aware MMKGC Agent with local memory enhancement. Unlike conventional models that rely on static embeddings, our agent establishes a unified **closed-loop cognitive architecture** to achieve dynamic adaptive reasoning. The core innovation of this framework lies in its transition from passive perception to an **interdependent cognitive cycle** that balances repre-

* Corresponding author.
E-mail addresses: yk2001@shu.edu.cn (K. Yang), wxz2017@shu.edu.cn (X. Wang), 632024230008@smail.nju.edu.cn (B. Lu), luoxf@shu.edu.cn (X. Luo), lchf@shu.edu.cn (C. Li), huangjq@shu.edu.cn (J. Huang).

https://doi.org/10.1016/j.eswa.2026.132001
Received 21 September 2025; Received in revised form 8 February 2026; Accepted 6 March 2026
Available online 11 March 2026

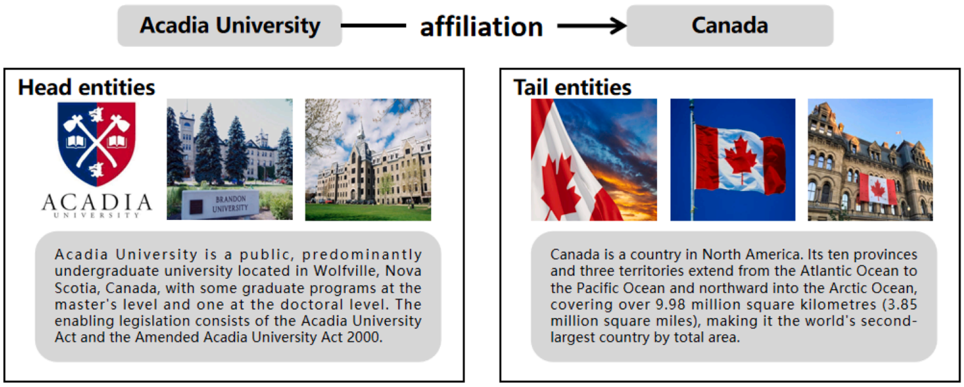


Fig. 1. Examples of multimodal link prediction.

sentational plasticity with semantic stability. While utilizing a Dual-Path Encoder to perceive basic global and local structures, the agent employs a **Learnable Residual Memory Pathway (LRMP)** to actively synthesize query-dependent memory states $\tilde{\mathbf{h}}''$. This mechanism provides the necessary plasticity for the agent to reconstruct entity representations based on specific relational contexts. Simultaneously, to counterbalance the representational drift inherent in such dynamic adaptation, we integrate a **Self-Alignment** Stabilization system that utilizes a redundancy-reducing Barlow Twins objective to enforce intrinsic coherence. This system acts as a semantic anchor, ensuring that the agent's adaptive reasoning remains grounded. Finally, the Reasoning & Action module translates these refined internal states into explicit behaviors, executing robust link prediction as the agent's external action (as shown in Fig. 1). This synergistic architecture empowers the agent to actively perceive structural cues, maintain context-aware memory, and purify semantic signals, thus overcoming the structural rigidity of static models.

In summary, our contributions are threefold:

- We introduce an **Agent-Based Paradigm for MMKGC** that fundamentally shifts the approach from static embedding matching to dynamic cognitive reasoning. Moving beyond simple component stacking, this architecture establishes a functional interdependency between perception, memory, and calibration to bridge global-local structural modeling with adaptive context-awareness.
- We propose a query-dependent **Learnable Residual Memory Pathway** that provides the agent with **representational plasticity**. By dynamically encoding fine-grained entity–relation interactions into short-term memory $\tilde{\mathbf{h}}''$, this mechanism breaks the static embedding bottleneck and enables the generation of context-sensitive semantics tailored to the current task.
- We design a **Semantic Self-Alignment** mechanism that serves as a **stability anchor**. By employing a redundancy-reduction objective, it enforces intrinsic representational coherence to counterbalance dynamic memory adaptation, thereby strengthening the agent's robustness against multimodal noise and prevent semantic drift during reasoning.

Extensive experiments on public MMKG benchmarks (Liu et al., 2019; Xu et al., 2022) demonstrate that our agent consistently outperforms over twenty state-of-the-art baselines, establishing a new paradigm for adaptive multimodal knowledge completion.

2. Related work

2.1. Multi-modal knowledge graph completion

Multi-modal knowledge graphs (MMKGs) (Chen et al., 2024) extend traditional knowledge graphs by incorporating diverse multimodal information, such as images, text descriptions, and other sensory data (Wang et al., 2023b), to enrich entity representations and enhance their contextual understanding. However, the inherent incompleteness of MMKGs poses significant challenges. Multimodal knowledge graph completion (MMKGC) aims to address this issue by leveraging both the structural information within triples and the multimodal characteristics of entities to predict missing links (Bordes et al., 2013).

Existing MMKGC methods primarily focus on integrating multimodal information into knowledge graph embedding (KGE) models (Bordes et al., 2013), which can be broadly categorized into three main approaches:

1. Multimodal Fusion Methods: These approaches unify multimodal information by extracting and integrating embeddings from different modalities. For example, IKRL (Xie et al., 2017) incorporated visual information into the TransE model (Bordes et al., 2013). TBKGC (Mousselly-Sergieh et al., 2018b) further extended this by integrating both visual and textual data. TransAE (Wang et al., 2019) employed autoencoders for multimodal fusion, while RSME (Wang et al., 2021) utilized gates to selectively incorporate relevant visual information. MyGO (Zhang et al., 2025) leveraged VisualBERT for deep multimodal feature extraction, and OTKGE (Cao et al., 2022a) utilized optimal transport for multimodal fusion. More recently, APKGC (Jian et al., 2025a) enhanced multimodal fusion through noise-augmented visual features and an attention penalty mechanism, which aims to suppress overly dominant modalities during representation learning. AMIT (Jian et al., 2025b) proposed an adaptive modality interaction transformer that models cross-modality relations through dynamic attention, enabling more flexible and fine-grained modality interactions. While these methods improve the integration of multimodal features, they typically rely on direct feature-level fusion and overlook deeper structural interactions within the graph.

2. Modal Ensemble Methods: These methods learn separate models for each modality and subsequently combine their predictions through ensemble learning techniques. For instance, IMF (Li et al., 2023) developed separate MMKGC models using distinct modality information and combined their outputs for joint decision-making. AdaMF-MAT (Zhang et al., 2024b) further introduced a modality-aware adaptive fusion mechanism through multi-head attention, enabling a more flexible ensemble of modality-specific predictors. Although these methods effectively utilize the unique features of each modality, they often treat each modality as an independent entity, limiting the exploration of modality interactions and the local structural semantics inherent in the knowledge graph.

3. Negative Sampling Enhanced Methods: These approaches focus on improving the negative sampling process in KGE models by incorporating multimodal information. MMRNS (Xu et al., 2022) enhanced the negative sampling process by utilizing modality-specific information, while MANS (Zhang et al., 2023) proposed a modality-aware negative sampling method to better align structural and multimodal features.

In addition, DSGNN (Li et al., 2025) introduced a decoupled semantic graph neural network to separately model relational semantics and entity semantics, which improves the robustness of the embedding process and benefits negative sampling by generating more consistent semantic spaces. Although these methods improve the quality of negative samples or semantic embeddings, they still predominantly focus on the structural aspects of the knowledge graph.

Overall, despite their diverse fusion strategies, existing methods generally treat MMKGC as a static embedding problem. Once trained, the representation of an entity remains fixed, making it difficult to adaptively capture fine-grained, query-specific semantics. In contrast to these feature-level or ensemble-based methods, our work introduces an architectural paradigm shift from static representation learning to dynamic cognitive reasoning. Instead of merely stacking perceptual modules, we propose an MMKGC-Agent that utilizes a LRMP to actively construct context-aware memory states. This enables the model to dynamically adjust its internal representations based on the specific query intent, fundamentally overcoming the "static bottleneck" of prior arts.

2.2. Contrastive learning

Contrastive learning (Chen et al., 2020b; Cheng et al., 2020) is a powerful technique for learning representations by pulling similar data points closer and pushing dissimilar ones apart. Instance discrimination methods, such as SimCLR (Chen et al., 2020b) and MoCo (He et al., 2020), treat each instance as a distinct class, learning embeddings by contrasting augmented views. Clustering-based methods like SwAV (Caron et al., 2020) generate pseudo-labels to improve semantic understanding. Supervised contrastive learning (SupCon) (Khosla et al., 2020) leverages labeled data to optimize class separation. However, most of these methods rely on negative sampling (InfoNCE loss), which can be computationally expensive and sensitive to batch sizes. In the context of our MMKGC-Agent, we diverge from standard instance discrimination. Instead, we adopt a redundancy-reduction approach inspired by Barlow Twins. We repurpose this objective as an internal semantic calibration mechanism. Rather than just distinguishing entities, our goal is to decorrelate the feature components within the multimodal embeddings, effectively "purifying" the signal from noise and modality-specific redundancy. This ensures that the agent's constructed memory is built upon distinct and semantically rich representations, enhancing robustness against modality dropout.

3. Task definition

We formalize the multimodal knowledge graph **environment** as $G = (E, R, T, V, C)$, where E denotes the set of entities, R the set of relations, and $T = \{(h, r, t) \mid h, t \in E, r \in R\}$ the observed triple set. Each entity $e \in E$ is associated with multimodal perceptual signals: a collection of visual inputs $V(e)$ and textual descriptions $C(e)$. This formulation captures both the logical structure and the perceptual diversity of the data, defining the external environment with which the agent interacts.

The objective of knowledge graph completion is reformulated here as a context-aware agentic decision process. Unlike static paradigms that map entities to fixed points in a vector space, our agent operates dynamically. Given an incomplete query (observation) such as $(h, r, ?)$, the agent does not merely retrieve a static embedding for h. Instead, it must actively construct a query-dependent memory state based on the interaction between h and r. Formally, the agent learns a policy parameterized by a scoring function $S(\tilde{\mathbf{h}}'', t) : \mathcal{M} \times E \to \mathbb{R}$, where $\tilde{\mathbf{h}}'' \in \mathcal{M}$ represents the dynamically generated memory state of the head entity under relation r, and t is a candidate tail entity. The goal is to maximize the plausibility score of the ground-truth action (entity selection) by refining internal memory states rather than just static features.

During the inference phase, the agent executes completion actions. For a query $(h, r, ?)$, the agent first transitions its internal state to reflect the specific semantic context of r and then evaluates all candidates $e \in E$ based on this updated state. The agent's success is measured by the ranking of the correct entity using standard metrics such as Mean Rank (MR), Mean Reciprocal Rank (MRR), and Hits@K. This evaluation reflects the agent's ability to perceive multimodal cues, construct context-aware memory, and perform robust reasoning in partial environments.

4. Methodology

In this section, we present the MMKGC-Agent, a unified cognitive framework designed to transcend the limitations of static representation learning. Distinct from traditional completion models that passively assign fixed embeddings to entities, the MMKGC-Agent operates as an active cognitive entity. The architecture is built upon a tightly coupled, closed-loop cognitive cycle that balances representational plasticity with semantic stability. It employs a sensory backbone to perceive multimodal signals, but crucially, it relies on a Learnable Residual Memory Pathway (LRMP) to actively construct and update short-term local memory states tailored to the current query. This process is further stabilized by an internal self-alignment mechanism that prevents representational drift during dynamic adaptation. By shifting the focus from static encoding to dynamic state construction, the agent is capable of adapting its internal semantics to varying relational contexts, thereby achieving robust reasoning even in incomplete or noisy environments. The overall structure of the MMKGC-Agent is shown in Fig. 2.

4.1. Perception module: Fine-grained global–local dual-path perception

The perception module (Hong et al., 2024; Wang et al., 2023a) enables the agent to convert multimodal observations into fine-grained semantic representations. Beyond existing multimodal encoders, our perception mechanism introduces token-level semantic decomposition and a unified global–local routing architecture, which together generate expressive representations tailored for knowledge-graph-centric reasoning. To comprehensively understand its environment, the agent performs two complementary levels of attentional processing: global structural perception and local neighborhood perception. These are implemented through a Fine-Grained Modality Tokenization process and a Dual-Path Encoder, providing both semantic atomization and effective global–local integration. This design ensures that the subsequent memory and reasoning modules are fed with high-quality, fine-grained evidence.

4.1.1. Fine-grained modality tokenization

To effectively capture fine-grained multimodal information, Perception Module propose a Fine-grained Modality Tokenization module designed to convert the original multimodal data of entities into discrete semantic tokens at a fine-grained level. These tokens act as semantic units, facilitating the learning of detailed and nuanced entity representations.We employ the tokenizers for image and textual modality respectively, denoted as T_{img} and T_{txt}, to generate visual tokens $v_{e,i}$ and textual tokens $t_{e,i}$ for entity e:

$$U_{\text{img}}(e) = \{v_{e,1}, v_{e,2}, \dots, v_{e,m_e}\} = T_{\text{img}}(V(e)) \tag{1}$$

$$U_{\text{txt}}(e) = \{t_{e,1}, t_{e,2}, \dots, t_{e,n_e}\} = T_{\text{txt}}(C(e)) \tag{2}$$

where m_e, n_e is the number of tokens for each modality, and we denote $U(e)$ as the collective token set for entity e. The text tokens are from the vocabulary of a language model , while the visual tokens are from the codebook of a pre-trained visual tokenizer. Notably, $V(e)$ might consist of multiple images, and we process each image to accumulate the tokens in $U_{\text{img}}(e)$. This tokenization process decomposes multimodal content into fine-grained semantic units, providing the agent with a detailed perceptual basis necessary for downstream context-sensitive reasoning and memory formation.

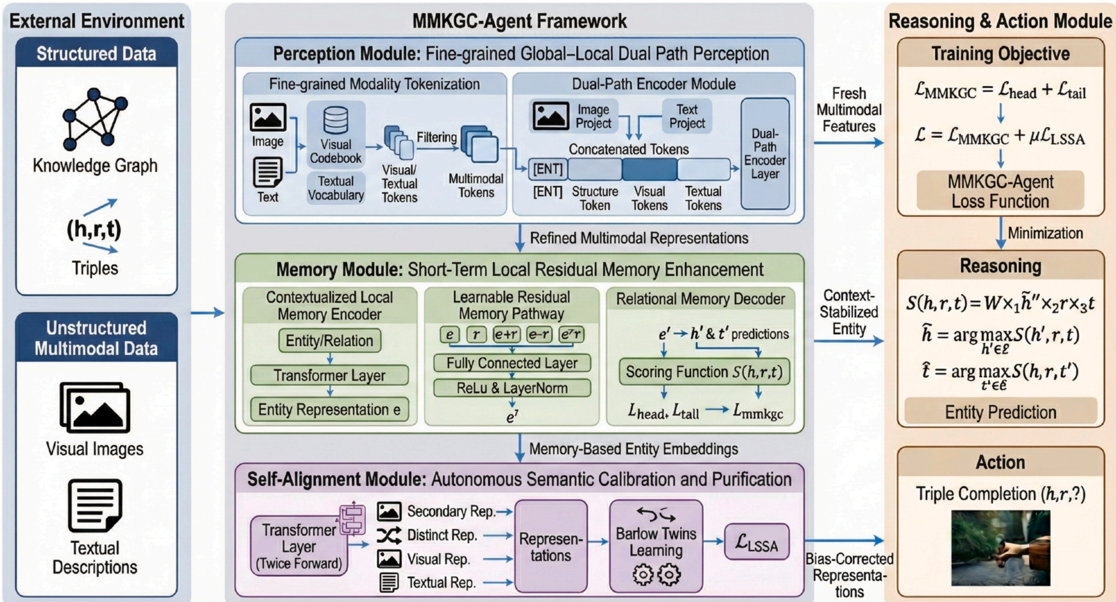


Fig. 2. The overview of our MMKGC-Agent framework. We mainly have four parts of new designs in MMKGC-Agent to integrate multimodal perception, enhance local memory, enable tool-augmented interaction, and perform adaptive reasoning.

4.1.2. Dual-path encoder module

To capture both cross-modal interactions and topological dependencies, the agent processes these tokens through a dual-stage pipeline. The multimodal tokens are first projected and concatenated into a transformer-ready sequence $X'(e)$, augmented with a learnable structural token s_e and a special classification token [ENT]:

$$X'(e) = \left([\mathrm{ENT}], s_e, \hat{v}_{e,1}, \dots, \hat{v}_{e,m}, \hat{t}_{e,1}, \dots, \hat{t}_{e,n}\right) \tag{3}$$

Here, $\hat{v}$ and $\hat{t}$ are projected features optimized via linear layers P_{img} and P_{txt} to align distributions. A standard Transformer encoder is then applied to model global dependencies across all modalities:

$$\mathbf{e} = \mathrm{Pooling}\left(\mathrm{Transformer}(X'(e))\right) \tag{4}$$

where $\mathbf{e}$ captures the global semantic context of the entity.

Instead of directly adopting this global representation as the entity embedding, the agent refines it through a dynamic graph attention convolutional layer to further capture local neighborhood interactions:

$$\mathbf{e}' = \mathrm{GAT}\left(\mathbf{e}, \mathrm{edge\_index}\right) \tag{5}$$

where edge_index denotes the structural edges of the multimodal knowledge graph. GAT($\cdot$) dynamically adjusts attention weights among entities according to their structural relationships, thereby performing adaptive neighborhood aggregation.

Distinct from conventional static methods that adopt the encoder output $\mathbf{e}$ as the final representation, our framework utilizes the neighborhood-refined $\mathbf{e}'$ merely as the initial static state. This state serves as the foundational “raw material” for the subsequent Memory Module to construct query-dependent dynamic representations.

4.2. Memory module: Short-term local residual memory enhancement

This module represents the cognitive core of the MMKGC-Agent. Upon acquiring the initial static state $\mathbf{e}'$ from the perception module, the agent transitions into the active memory construction phase. We posit that a static embedding $\mathbf{e}'$, no matter how refined, struggles to effectively capture the distinct semantic roles an entity may play across different contexts. Therefore, unlike traditional approaches that directly utilize $\mathbf{e}$ for inference, our agent generates a dynamic, query-specific memory state $\tilde{\mathbf{h}}''$. This state is not a fixed parameter but a transient representation conditioned explicitly on the current query $(h, r, ?)$. By leveraging the sensory input $\mathbf{e}'$, the memory component selectively highlights neighborhood-level semantics relevant to the relation r while retaining global context. The memory construction process is orchestrated through three synergistic stages: Contextualized Local Memory Encoding, Learnable Residual Memory Interaction, and Relational Memory Decoding.

4.2.1. Contextualized local memory encoder

To initiate the memory construction process, the agent first maps the retrieved perceptual features into a query-specific latent space. This step acts as a **contextual trigger**, activating relevant semantic signals while suppressing irrelevant noise. For a specific query $(h, r, ?)$, the retrieval is governed by a contextual token:

$$\tilde{\mathbf{h}} = \mathrm{Transformer}\left([\mathrm{CXT}], \mathbf{h}, \mathbf{r}\right) \tag{6}$$

Here, $\mathbf{h}$ is the entity embedding supplied by the perception backbone, $\mathbf{r}$ is the relation embedding, and [CXT] serves as a specialized prompt token to focus attention. Crucially, the resulting $\tilde{\mathbf{h}}$ is no longer a generic entity representation; it captures the **initial memory state** of the entity specifically conditioned on the interaction with relation r, marking the transition from passive perception to active memory formation.

4.2.2. Learnable residual memory pathway

A single pass of contextual encoding is insufficient to capture intricate structural dependencies. To address this, we introduce the LRMP, the engine of the agent’s dynamic reasoning capability. Rather than merely storing features, LRMP equips the agent with a constructive reasoning mechanism. It explicitly models fundamental entity-relation interaction patterns—such as translation, composition, and correlation—to enrich the short-term memory.

Concretely, LRMP performs a diversified set of explicit interactions between the memory state $\tilde{\mathbf{h}}$ and the relation $\mathbf{r}$:

$$\mathbf{c} = \mathrm{CrossInteractLayer}(\tilde{\mathbf{h}}, \mathbf{r}) = \mathrm{Dropout}\left(\mathrm{Linear}\left([\tilde{\mathbf{h}}; \mathbf{r}; \tilde{\mathbf{h}} + \mathbf{r}; \tilde{\mathbf{h}} - \mathbf{r}; \tilde{\mathbf{h}} \odot \mathbf{r}]\right)\right) \tag{7}$$

These operations are not arbitrary; they correspond to distinct logical reasoning inductive biases. Specifically, $\tilde{\mathbf{h}} - \mathbf{r}$ captures translational semantics, $\tilde{\mathbf{h}} \odot \mathbf{r}$ captures multiplicative interactions, and $\tilde{\mathbf{h}} + \mathbf{r}$ models additive composition. By fusing these distinct memory slices, the agent reconstructs a comprehensive view of the entity's role in the current triple.

The agent then integrates these interactive features back into the memory stream via a residual connection:

$$\tilde{\mathbf{h}}' = \mathbf{c} + \tilde{\mathbf{h}} \tag{8}$$

followed by non-linear transformation and normalization:

$$\tilde{\mathbf{h}}'' = \mathrm{ReLU}\Big(\mathrm{LayerNorm}\big(\mathrm{Linear}(\tilde{\mathbf{h}}')\big)\Big) \tag{9}$$

This pathway ensures that the local memory is not only retained but continually **intensified**. Through this process, weak structural cues are amplified into highly discriminative memory representations ($\tilde{\mathbf{h}}''$) that are robust against noise and missing modalities.

4.2.3. Relational memory decoder

In the final phase, the agent translates its constructed memory state into explicit behavior. The Relational Memory Decoder serves as the decision-making unit, computing the plausibility of candidate triples based on the dynamic memory $\tilde{\mathbf{h}}''$:

$$S(h, r, t) = \mathcal{W} \times_1 \tilde{\mathbf{h}}'' \times_2 \mathbf{r} \times_3 \mathbf{t}, \tag{10}$$

where $\mathcal{W} \in \mathbb{R}^{d\times d\times d}$ is the learnable core tensor, and $\times_n$ denotes the mode-n tensor product. It is vital to underscore that this score reflects dynamic inference: unlike static models that match fixed embeddings, the agent evaluates candidates based on the $\tilde{\mathbf{h}}''$ it just actively constructed for this specific query.

During training, to endow the agent with bidirectional reasoning capabilities, we model the conditional distributions $p(t \mid h, r)$ and $p(h \mid r, t)$. We employ a cross-entropy objective over the entity set:

$$\mathcal{L}_{\text{head}} = -\sum_{(h,r,t)\in\mathcal{T}} \log \frac{\exp S(h,r,t)}{\sum_{t'\in\mathcal{E}} \exp S(h,r,t')}, \tag{11}$$

$$\mathcal{L}_{\text{tail}} = -\sum_{(h,r,t)\in\mathcal{T}} \log \frac{\exp S(h,r,t)}{\sum_{h'\in\mathcal{E}} \exp S(h',r,t)}. \tag{12}$$

The global training objective is defined as:

$$\mathcal{L}_{\text{MMKGC}} = \mathcal{L}_{\text{head}} + \mathcal{L}_{\text{tail}}, \tag{13}$$

ensuring the agent maintains consistent reasoning logic across diverse completion scenarios.

4.3. Self-alignment module: Autonomous semantic calibration and purification

While the Memory Module (Section 4.2) empowers the agent to dynamically adapt its state to varying query contexts, this flexibility introduces a potential risk where entity representations might drift or become over-specialized to specific relations. To counterbalance this dynamic adaptation, we introduce the **Learnable Semantic Self-Alignment** mechanism as a parallel stabilization system. Its primary function is to enforce intrinsic semantic coherence across the agent's internal states by acting as a semantic anchor that ensures the learned representations remain discriminative and robust during dynamic reasoning.

To achieve this, the agent constructs a candidate set of complementary semantic views for each entity e through different pooling and slicing operations on the multimodal token sequence. Specifically, we define four heterogeneous views to capture diverse semantic facets: (1) the classification token state $\mathbf{e}_{\text{ent}}$ which aggregates holistic entity information, (2) the global mean representation $\mathbf{e}_{\text{avg}}$ derived from all tokens, (3) the visual-specific mean $\mathbf{e}_{\text{vis}}$ which focuses on image features, and (4) the textual-specific mean $\mathbf{e}_{\text{txt}}$ which emphasizes descriptive content. To further enforce robustness against structural perturbations, these views are augmented with a learnable latent token. Collectively, these views form the alignment set:

$$\mathcal{C}(e) = \left\{\mathbf{e}_{\text{ent}}, \mathbf{e}_{\text{avg}}, \mathbf{e}_{\text{vis}}, \mathbf{e}_{\text{txt}}\right\} \tag{14}$$

where each view captures a unique fine-grained nuance of the entity.

The goal of LSSA is to ensure these diverse views remain consistent while minimizing informational redundancy. Unlike traditional contrastive paradigms that rely on negative sampling, LSSA adopts a redundancy-reduction strategy inspired by the Barlow Twins objective. This approach focuses solely on the cross-correlation matrix of the internal representations to ensure that alignment is achieved through intrinsic coherence rather than external opposition. The semantic-alignment objective for a pair of views (e, e') is formulated as:

$$\mathcal{L}_{\text{Semantic}}(e, e') = \sum_{i=1}^{D} (1 - c_{i,i})^2 + \lambda \sum_{i\neq j} c_{i,j}^2 \tag{15}$$

where $c_{i,j}$ measures the cross-correlation between the ith and jth dimensions of the embeddings from views e and e', while λ regulates the trade-off between the invariance term and the redundancy-reduction term. The invariance term forces diagonal elements to 1 to ensure that diverse views point to the same underlying concept, while the redundancy-reduction term drives off-diagonal elements to 0 to disentangle the semantic axes.

By minimizing this objective across all pairs in $\mathcal{C}(e)$, the agent achieves concurrent semantic purification:

$$\mathcal{L}_{\text{LSSA}} = \frac{1}{|\mathcal{C}(e)|} \sum_{e'\in\mathcal{C}(e)} \mathcal{L}_{\text{Semantic}}(e, e') \tag{16}$$

In practice, the LSSA module is integrated into the framework as an auxiliary loss term that is optimized jointly with the primary reasoning objective. To maintain computational efficiency, the agent employs a stochastic sub-sampling strategy where a subset of entities is randomly selected in each batch to compute the alignment loss. This simultaneous optimization ensures that the Memory Module explores query-specific local interactions while the Self-Alignment Module preserves the purity of the underlying entity semantics. This synergistic design significantly enhances the overall robustness of the framework against multimodal noise and representation drift.

4.4. Reasoning & action module: From cognitive states to explicit behaviour

Following perception, semantic calibration, and memory construction, the agent enters the final Reasoning & Action phase. This module does not operate in isolation; rather, it serves as the actuation interface that translates the agent's internal cognitive states ($\tilde{\mathbf{h}}''$) into explicit interactions with the external environment.

For a given query $(h, r, ?)$ or $(?, r, t)$, the agent leverages its dynamically constructed short-term memory $\tilde{\mathbf{h}}''$—rather than the static retrieval used in conventional models—to execute the inference policy. The scoring function, realized via the Relational Memory Decoder (Section 4.2), acts as the reasoning engine:

$$\hat{h} = \arg\max_{h'\in\mathcal{E}} S(h', r, t), \quad \hat{t} = \arg\max_{t'\in\mathcal{E}} S(h, r, t') \tag{17}$$

where $S(\cdot)$ quantifies the compatibility between the current memory state and candidate entities. This procedure embodies adaptive cognition: the agent's decision is not based on a fixed entity feature, but on a transient state specifically synthesized to answer the current query.

To ensure the agent develops both external adaptability and internal coherence, the framework is optimized via a composite objective:

$$\mathcal{L} = \mathcal{L}_{\text{MMKGC}} + \mu\, \mathcal{L}_{\text{LSSA}} \tag{18}$$

Here, $\mathcal{L}_{\text{MMKGC}}$ drives the agent to maximize prediction accuracy, while $\mathcal{L}_{\text{LSSA}}$ acts as an intrinsic regularizer, ensuring that the representations used for memory construction remain semantically distinct and robust against semantic drift.

In summary, the MMKGC-Agent is not a mere stack of deep learning layers, but a functional system operating under a four-stage cognitive cycle. Each stage is strictly interdependent, forming a unified paradigm for dynamic reasoning:

- **Perception Module:** $o_t \to \mathbf{e}'$. The hybrid backbone extracts raw structural and multimodal signals, serving as the static foundation.
- **Memory Module:** $\mathbf{e}', r \to \tilde{\mathbf{e}}''$. The LRMP dynamically transforms the static input into a query-dependent memory state $\tilde{\mathbf{e}}''$, adapting the internal representation to the specific relation context.
- **Self-Alignment Module:** $\mathbf{e}' \to$ Stabilized $\mathbf{e}'$. Simultaneously, the LSSA mechanism enforces intrinsic representational coherence to counterbalance dynamic adaptation, acting as a semantic anchor to prevent contextual overfitting.
- **Reasoning & Action Module:** $\tilde{\mathbf{e}}'' \to a_t$. The agent executes the link prediction action a_t based on the local memory state, completing the cognitive loop.

This closed-loop design ensures that the agent moves beyond traditional feature fusion, achieving a system where perception serves memory, calibration stabilizes memory, and action manifests memory.

4.5. Formalizing the cognitive advantages of closed-loop architecture

To strengthen the conceptual rigor of our agent-based formulation, we formalize why the MMKGC-Agent yields cognitive advantages and supports transfer across heterogeneous reasoning scenarios. We model the MMKGC-Agent as a discrete-time, partially observable dynamical system with an internal memory update operator. At time step t, the agent observes multimodal input o_t (derived from images, text, and graph neighborhoods), maintains internal short-term memory $m_t \in \mathbb{R}^d$, and produces action a_t (prediction or scoring decision). The closed-loop update is written compactly as:

$$m_{t+1} = \Phi(m_t, o_t, a_t), a_t \sim \pi(\cdot \mid m_t, o_t), \tag{19}$$

where Φ denotes the memory update operator implemented by the LRMP (including residual connections, normalization, and nonlinearities), and π denotes the reasoning policy implemented by the Relational Memory Decoder. The alignment module (LSSA) introduces a semantic projection Ψ that regularizes internal feature dimensions; we will make its effect explicit in what follows.

Assume the memory operator $\Phi(\cdot, o, a)$ is locally Lipschitz continuous with respect to its memory argument. In our architecture, the integration of residual connections, layer normalization, and bounded nonlinearities (such as ReLU) effectively enforces a bounded gain property. Specifically, we consider typical observation sets $\mathcal{O}$ and policy-generated actions a, such that there exists a contraction constant $0 \le \lambda < 1$. Under this condition, for any memory states m_1 and m_2, the operator satisfies:

$$\|\Phi(m_1, o, a) - \Phi(m_2, o, a)\| \le \lambda \|m_1 - m_2\|. \tag{20}$$

By the Banach fixed-point theorem, this property implies geometric convergence to a unique, query-dependent fixed point $m^\star(o, a)$. Starting from any initial state m_0, the iterative update $m_{t+1} = \Phi(m_t, o, a)$ converges as:

$$\|m_t - m^\star\| \le \lambda^t \|m_0 - m^\star\|. \tag{21}$$

This formalization confirms the intuition that when LRMP is designed with bounded gain, it concentrates memory updates and prevents divergent oscillations. Crucially, since $m^\star$ is explicitly conditioned on the observation o and action a through the query $(h, r, ?)$, the agent achieves query-sensitive semantics while maintaining structural stability.

We next characterize robustness against perturbations (observation noise, missing modalities, or stochastic optimization noise). Model the memory update with additive perturbation:

$$m_{t+1} = \Phi(m_t, o_t, a_t) + \eta_t, \tag{22}$$

where η_t is a bounded perturbation ($\|\eta_t\| \le \epsilon$). Then:

With contraction constant $\lambda < 1$ and bounded perturbation $\sup_t \|\eta_t\| \le \epsilon$, the memory error relative to the nominal fixed point is uniformly bounded:

$$\|m_t - m^\star\| \le \lambda^t \|m_0 - m^\star\| + \frac{1}{1-\lambda}\epsilon. \tag{23}$$

The first term shows geometric forgetting of initial conditions; the second term shows that perturbations are not amplified unboundedly, but contribute only $\mathcal{O}(\frac{\epsilon}{1-\lambda})$ steady-state error. Thus, contraction (small λ) reduces sensitivity to noisy inputs and missing modalities. In practice, LRMP's residual + normalization layers help keep λ in the < 1 regime, thereby guaranteeing the noise-suppressing capability of memory trajectories.

We now connect memory convergence to the reliability of the scoring function used for link prediction. Assume the relational memory decoder $S(\cdot, r, t)$ (for scoring candidate entities) is L_S-Lipschitz in its memory argument:

$$|S(m_1, r, t) - S(m_2, r, t)| \le L_S \|m_1 - m_2\|. \tag{24}$$

Combining with previous results, we obtain a quantitative bound on scoring errors due to incomplete convergence or noise:

$$|S(m_t, r, t) - S(m^\star, r, t)| \le L_S \left(\lambda^t \|m_0 - m^\star\| + \frac{\epsilon}{1-\lambda} \right). \tag{25}$$

Thus, contraction improves both the speed at which scores approach their steady-state values and limits the steady-state scoring error induced by perturbations.

Finally, we formalize how Learnable Semantic Self-Alignment (LSSA) acts as a stabilizing regularizer that reduces representation drift and conditions the decoder. Let C denote the empirical covariance matrix of memory representations used by the decoder across views; LSSA minimizes a redundancy reduction penalty of the form:

$$\mathcal{L}_{\text{LSSA}} \propto \sum_{i \neq j} c_{ij}^2 + \sum_i (1 - c_{ii})^2, \tag{26}$$

which drives C toward an (approximately) diagonal matrix with unit variance. This yields two important consequences.

First, by reducing off-diagonal covariances, LSSA decreases multicollinearity among feature axes. For linear or locally linear decoders, the condition number of the design matrix is improved; specifically, the condition number $\kappa(C)$ is reduced, which tightens standard least-squares generalization bounds and improves numerical stability of parameter updates. Second, decorrelated feature axes limit the sensitivity of downstream mappings: if the decoder's operator norm depends on the maximum eigenvalue $\lambda_{\max}(C)$, then LSSA (by controlling cross-correlations) prevents pathological growth of $\lambda_{\max}$, thereby helping keep the decoder Lipschitz constant L_S moderate.

We can make these relationships explicit in a simplified linearized setting. Consider the linearized decoder $S(m) = w^\top m$, and let $\tilde{C} = \mathbb{E}[(m - m^\star)(m - m^\star)^\top]$ denote the empirical covariance of noisy memory estimates. Under a small-noise Gaussian approximation, the variance of score $S(m)$ is $w^\top \tilde{C} w$. Minimizing off-diagonal terms of $\tilde{C}$ reduces this variance for any fixed w, thereby improving robustness of ranking scores under perturbations. More generally, LSSA reduces the effective degrees of freedom that the decoder must manage to distinguish entities, which decreases sample complexity constants in the decoder's statistical learning bounds.

The above formalization translates qualitative intuitions into quantitative statements: under benign but realistic architectural constraints, the closed-loop MMKGC-Agent possesses query-dependent fixed points, converges to them geometrically, propagates input perturbations in a controlled (non-amplifying) manner, and benefits from an alignment penalty that reduces variance and conditions downstream inference. Although we do not claim tight finite-sample generalization bounds for the full nonlinear system, these results provide principled theoretical support for the empirical gains observed in Section 5, and demonstrate why closed-loop cognitive cycles can reliably improve robustness and transfer in multimodal knowledge completion tasks.

5. Experiments

In this section, we present comprehensive experiments to validate the effectiveness of the proposed MMKGC-Agent with Local Memory

Table 1
Statistical information of the datasets.

Dataset	$\lvert\mathcal{E}\rvert$	$\lvert\mathcal{R}\rvert$	#Train	#Valid	#Test	$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{D}\rvert$
DB15K	12,842	279	79,222	9902	9904	12,818	12,842
MKG-W	15,000	169	34,196	4276	4274	14,463	14,123
MKG-Y	15,000	28	21,110	2665	2665	14,244	12,826
KVC16K	16,015	4	180,190	22,523	22,252	14,822	14,822

Enhancement. Rather than simply reporting performance scores, the experiments are framed as controlled evaluations of the agent's cognitive abilities—its capacity to perceive, enhance local memory, self-align, and reason within heterogeneous multimodal environments. The study is organized around the following research questions (RQs), which reflect both global–local structural reasoning and fine-grained semantic alignment:

- **RQ1:** Can the agent, through its unified cognitive framework, outperform existing baseline methods in multimodal KG completion?
- **RQ2:** How essential is each cognitive component (perception, memory, self-supervision, reasoning) for sustaining the agent's capabilities?
- **RQ3:** To what extent does global–local dual-path perception improve the robustness of structural understanding?
- **RQ4:** How do memory-enhanced residual interactions contribute to adaptive entity representation under sparse structures?
- **RQ5:** Does self-supervised alignment via contrastive learning enhance the semantic purity of multimodal embeddings?
- **RQ6:** How sensitive is the agent's behavior to hyper-parameter variations, and what does this imply for adaptability?
- **RQ7:** Compared to static models, how does the agent balance cognitive effectiveness with computational efficiency?
- **RQ8:** How well can the agent generalize to multimodal link prediction tasks beyond unimodal or single-view scenarios, leveraging its memory-enhanced reasoning?

5.1. Experiment settings

5.1.1. Datasets

We regard benchmark datasets as external environments that the agent needs to sense and interact with. Four multimodal knowledge graph completion benchmarks are adopted: DB15K (Chen et al., 2020a), MKG-W (Zhang et al., 2025), MKG-Y and KVC16K (Zhang et al., 2024a). DB15K is derived from DBpedia, while MKG-W and MKG-Y are subsets of Wikidata and YAGO, respectively, both providing structured triples enriched with textual descriptions and entity-associated images, thereby simulating environments with heterogeneous multimodal cues. KVC16K is a large-scale multimodal knowledge graph focusing on visual–conceptual associations, containing a small number of relations but a high proportion of multimodal entities, which allows for evaluating the agent's performance in low-relational yet visually grounded scenarios. These datasets differ in scale, density, and modality balance, enabling us to comprehensively examine the agent's adaptability under both sparse and richly multimodal conditions. Detailed statistics are shown in Table 1, where $|\mathcal{E}|$ and $|\mathcal{R}|$ denote the numbers of entities and relations, respectively, while $|\mathcal{V}|$ and $|\mathcal{D}|$ represent the coverage of visual and textual descriptions.

5.1.2. Task and evaluation protocols

The evaluation focuses on the **link prediction** task, which directly tests whether the agent can complete missing information in multimodal environments. Given an incomplete triple $(h, r, ?)$ or $(?, r, t)$, the agent must infer the most plausible entity.

Following standard practice, we employ Mean Reciprocal Rank (MRR) and Hit@K ($K = 1, 3, 10$) as evaluation metrics. These metrics capture not only the correctness but also the ranking quality of the agent's predictions. To ensure fair comparison, the *filtered* setting is applied, where candidate triples already seen in training are removed. The final scores are averaged over both head and tail prediction. Formally:

$$\mathrm{MRR} = \frac{1}{|\mathcal{T}|}\sum_{i=1}^{|\mathcal{T}|}\left(\frac{1}{p_{h,i}} + \frac{1}{p_{t,i}}\right) \tag{27}$$

$$\mathrm{Hit@K} = \frac{1}{|\mathcal{T}|}\sum_{i=1}^{|\mathcal{T}|}\Big(\mathbb{I}(p_{h,i} \le K) + \mathbb{I}(p_{t,i} \le K)\Big) \tag{28}$$

where $p_{h,i}$ and $p_{t,i}$ are the results of head prediction and tail prediction, and $\mathcal{T}$ is the triple set.

5.1.3. Baseline methods

To position the agent's performance in context, we compare it with 25 state-of-the-art methods, categorized as follows:

(1). **Uni-modal KGC methods**: TransE, DistMult, ComplEx, RotatE, PairRE, GC-OTE, and Tucker.These conventional methods only consider the structure information in their model design.

(2). **MMKGC methods**: IKRL, TBKGC, TransAE,MMKRL, RSME, VBKGC, OTKGE, IMF, QEB, VISTA, AdaMF, AdaMF-MAT, APKGC, AMIT, and DSGNN. These methods consider image and textual information in the MMKGC models and achieve multimodal fusion with different settings.

(3). **Negative sampling methods**: MANS, and MMRNS. These methods utilize the multi-modal information in the MMKGs to generate high-quality negative samples for training.

(4). **Fine-Grained methods**: MyGO. This method tokenizes the modality data into fine-grained multi-modal tokens and pioneers a novel MMKGC architecture to hierarchically model the cross-modal entity representation.

Although this work is philosophically inspired by recent advances in general-purpose AI agents, such as DS-Agent (Guo et al., 2024), COPPER (Bo et al., 2024), and AutoMLAgent (Trirat et al., 2025), it is important to emphasize that these frameworks are primarily designed for high-level reasoning tasks, including automated planning, decision-making, or code generation based on large language model (LLM) APIs. Their observation spaces and action mechanisms are tightly coupled with linguistic or programming environments, and thus lack the structural constraint modeling capabilities as well as multimodal semantic alignment and fusion mechanisms that are essential for large-scale knowledge graph completion. As a result, there exists a fundamental mismatch in terms of task objectives, data modalities, and model inductive biases, rendering direct empirical comparisons with such general-purpose agents methodologically inappropriate.

In contrast to conventional knowledge graph completion models, the goal of this work is not to directly adopt or replicate existing general-purpose agent frameworks. Instead, we are the first to systematically introduce the agent paradigm into the domain of multimodal knowledge graph completion (MMKGC), where autonomous decision-making and multimodal perception are leveraged to enable dynamic and adaptive optimization of the completion process. In line with this objective, we evaluate the proposed MMKGC-Agent against state-of-the-art structural baselines and multimodal baselines that are specifically designed and optimized for knowledge graph completion, in order to provide a fair and effective assessment of its advantages.

Some other approaches, such as KG-BERT that fine-tunes large-scale pretrained language models, are orthogonal to our design and thus omitted.

5.1.4. Implementation details

The proposed MMKGC-Agent is implemented in PyTorch, with its training pipeline designed to simulate adaptive cognitive interactions. For multimodal perception, we tokenize textual and visual inputs following the fine-grained strategy of MyGO, thereby transforming raw

Table 2
The main MMKGC results on DB15K, MKG-W, MKG-Y and KVC16K datasets. We list the type of method category (Uni-modal/Multi-modal/Negative Sampling/Agent-based) considered by each method in the table. The best results are marked as **bold** and the second-best results are underlined. The percentages in the last row indicate the **relative improvement** over the best baseline method ($\frac{\text{Ours}-\text{Baseline}}{\text{Baseline}}$).

Model	Method Category	DB15K				MKG-W				MKG-Y				KVC16K			
		MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10
TransE	Uni-modal	24.86	12.78	31.48	47.07	29.19	21.06	33.20	44.23	30.73	23.45	35.18	43.37	8.54	0.64	10.33	23.42
DistMult	Uni-modal	23.03	14.78	26.28	39.59	20.99	15.93	22.28	30.86	25.04	19.33	27.80	35.95	6.37	3.03	6.55	12.61
ComplEx	Uni-modal	27.48	18.37	31.57	45.37	24.93	19.09	26.69	36.73	28.71	22.26	32.12	40.93	12.85	7.48	12.37	23.18
RotatE	Uni-modal	29.28	17.87	36.12	49.66	33.67	26.80	36.68	46.73	34.95	29.10	38.35	45.30	14.33	8.25	15.97	26.17
PairRE	Uni-modal	31.13	21.62	35.91	49.30	34.40	28.24	36.71	46.04	32.01	25.53	35.83	43.89	12.17	6.05	13.30	24.19
GC-OTE	Uni-modal	31.85	22.11	36.52	51.18	33.92	26.55	35.96	46.05	32.95	26.77	36.44	44.08	13.09	7.52	14.47	25.83
TuckER	Uni-modal	33.86	25.33	37.91	50.38	30.39	24.44	32.91	41.25	29.12	23.50	34.15	40.78	9.87	5.68	12.45	21.12
IKRL	Multi-modal	26.82	14.09	34.93	49.09	32.36	26.11	34.75	44.07	33.22	30.37	34.28	38.26	11.11	5.42	12.45	22.39
TBKGC	Multi-modal	28.40	15.61	37.03	49.86	31.48	25.31	33.98	43.24	33.99	30.47	35.27	40.07	5.39	0.35	6.94	15.52
TransAE	Multi-modal	28.09	21.25	31.17	41.17	30.00	21.23	34.91	44.72	28.10	25.31	29.10	33.03	10.81	5.31	11.13	21.89
MMKRL	Multi-modal	26.81	13.85	35.07	49.39	30.10	22.16	34.09	44.69	36.81	31.66	39.79	45.31	8.78	3.89	8.91	18.34
RSME	Multi-modal	29.76	24.15	32.12	40.29	29.23	23.36	31.97	40.43	34.44	31.78	36.07	39.09	12.31	7.14	14.24	22.05
VBKGC	Multi-modal	30.61	19.75	37.18	49.44	30.61	24.91	33.01	40.88	37.04	33.76	38.75	42.30	14.66	8.28	17.38	27.04
OTKGE	Multi-modal	23.86	18.45	25.89	34.23	34.36	28.85	36.25	44.88	34.36	28.85	36.25	44.88	8.77	5.01	8.54	15.55
IMF	Multi-modal	32.25	24.2	36.00	48.19	34.50	28.77	36.62	45.44	35.79	32.95	37.14	40.63	12.01	7.42	12.49	21.01
QEB	Multi-modal	28.18	14.82	36.67	51.55	32.38	25.47	35.06	45.32	34.37	29.49	36.95	42.32	12.06	5.57	14.48	25.01
VISTA	Multi-modal	30.42	22.49	33.56	45.94	32.91	26.12	35.38	45.61	30.45	24.87	32.39	41.53	11.89	6.97	12.87	21.27
AdaMF	Multi-modal	32.51	21.31	39.67	51.59	34.27	27.21	37.86	47.21	38.06	33.49	40.44	45.48	14.12	7.05	13.35	26.66
AdaMF-MAT	Multi-modal	34.88	25.07	40.77	51.63	35.38	28.88	38.18	47.65	38.57	34.34	40.59	45.76	15.22	9.64	18.05	28.88
APKGC	Multi-modal	35.88	27.73	40.81	52.01	35.56	29.01	38.20	48.56	36.12	28.76	37.15	44.66	15.82	9.86	17.99	28.71
AMIT	Multi-modal	36.17	28.12	40.05	51.04	35.51	29.22	37.88	46.65	35.78	29.16	38.88	44.74	13.24	8.92	16.48	28.44
DSGNN	Multi-modal	35.72	29.07	39.32	50.98	35.41	28.31	37.83	47.23	35.53	27.21	37.84	45.12	13.75	8.49	17.54	28.76
MANS	Negative Sampling	28.82	16.87	36.58	49.26	30.88	24.89	33.63	41.78	29.03	25.25	31.35	34.49	10.42	5.21	12.15	20.45
MMRNS	Negative Sampling	32.68	23.01	37.86	51.01	35.03	28.59	37.49	47.47	35.93	30.53	39.07	45.47	13.31	7.51	14.46	24.68
MyGO	Fine-Grained	37.20	29.61	40.84	51.65	35.74	29.34	38.10	47.51	38.44	35.01	39.84	44.19	15.84	9.78	18.05	28.67
MMKGC-Agent	Agent-based	**38.32** **+3.0%**	**30.38** **+2.6%**	**42.44** **+3.9%**	**53.25** **+2.4%**	**37.16** **+4.0%**	**30.95** **+5.5%**	**39.83** **+4.3%**	**48.84** **+0.6%**	**39.43** **+2.2%**	**37.05** **+5.8%**	**41.56** **+2.4%**	**45.88** **+0.2%**	**16.67** **+5.2%**	**10.34** **+4.9%**	**18.85** **+4.4%**	**29.52** **+2.2%**

environmental cues into semantic atoms. For structural perception, the agent employs a Transformer encoder to capture global dependencies, followed by a GATv2-based neighbourhood aggregator to refine local interactions. Training is conducted for 1800 epochs with a batch size of 1024 and an embedding dimension of 256, reflecting the balance between cognitive capacity and computational efficiency. The contrastive self-alignment weight μ is tuned via a grid search from the set $1000, 100, 10, 1$, which regulates the strength of self-supervision in intrinsic stabilization constraint. Optimisation is carried out using Adam, with the learning rate selected from $\exp(-4)$, $2\exp(-4)$, …, $\exp(-3)$, ensuring stable convergence of the agent's cognitive loop. For baseline models, we either adopt their official implementations or re-implement them using the OpenKE framework, ensuring consistency in environmental conditions. All experiments are conducted on a Linux server equipped with an NVIDIA A100 GPU; training and evaluation across datasets typically take 3–5 hours. This setup guarantees that observed differences stem from cognitive design rather than computational artefacts.

5.2. Main results (RQ1)

The primary results for the MMKGC-Agent are presented in Table 2, which benchmarks its performance against 25 representative methods across four multimodal knowledge-graph completion datasets including DB15K, MKG-W, MKG-Y, and KVC16K.

First, the MMKGC-Agent demonstrates consistent performance gains across all ranking metrics on all four datasets. On DB15K, our model attains an MRR of 38.32, representing an absolute increase of 1.12 over the best non-agent baseline, which translates to a $+3.0\%$ relative improvement. Similarly, we observe relative MRR improvements of $+3.8\%$, $+2.2\%$, and $+5.2\%$ on MKG-W, MKG-Y, and KVC16K, respectively. Notably, these improvements extend beyond averaged metrics. The most significant relative gains are typically observed in $Hit@1$ and $Hit@3$. This pattern indicates that the MMKGC-Agent positions the ground-truth entity at the top of the ranking more frequently. For instance, the $+5.8\%$ increase in $Hit@1$ on MKG-Y underscores a significant boost in top-1 precision, which is critical for practical retrieval and downstream decision-making tasks.

We attribute these gains to the synergistic interplay within the closed-loop architecture. Unlike methods that rely on static embeddings or single-pass fusion, our agent utilizes the functional interdependency between the LRMP and the self-alignment mechanism. The LRMP provides the necessary representational plasticity to capture query-dependent cues that are often lost in static models. When a query necessitates attending to specific visual attributes or local neighborhoods, the memory module synthesizes a transient state that is simultaneously stabilized by the self-alignment anchor. This ensures that the agent amplifies relevant signals without introducing noise. Consequently, improvements are more pronounced in $Hit@1$ and MRR than in broader metrics like $Hit@10$, reflecting the agent's capacity to produce high-confidence, precise predictions through deeper cognitive deliberation.

Performance variations across datasets further validate the effectiveness of this non-trivial architectural integration. Specifically, the agent achieves its most substantial gains on KVC16K and MKG-W, which are datasets characterized by rich yet potentially noisy multimodal signals. This advantage confirms that the self-alignment module does not function as a mere post-processing step but as an intrinsic regularizer that enables the memory module to operate effectively in "noisy" environments. In the case of MKG-Y, where differentiating fine-grained semantic roles is the dominant challenge, the distinct improvement in $Hit@1$ suggests that the agent's dynamic state construction successfully resolves the semantic ambiguity that often limits static baselines. Furthermore, even on DB15K, the agent maintains its lead-

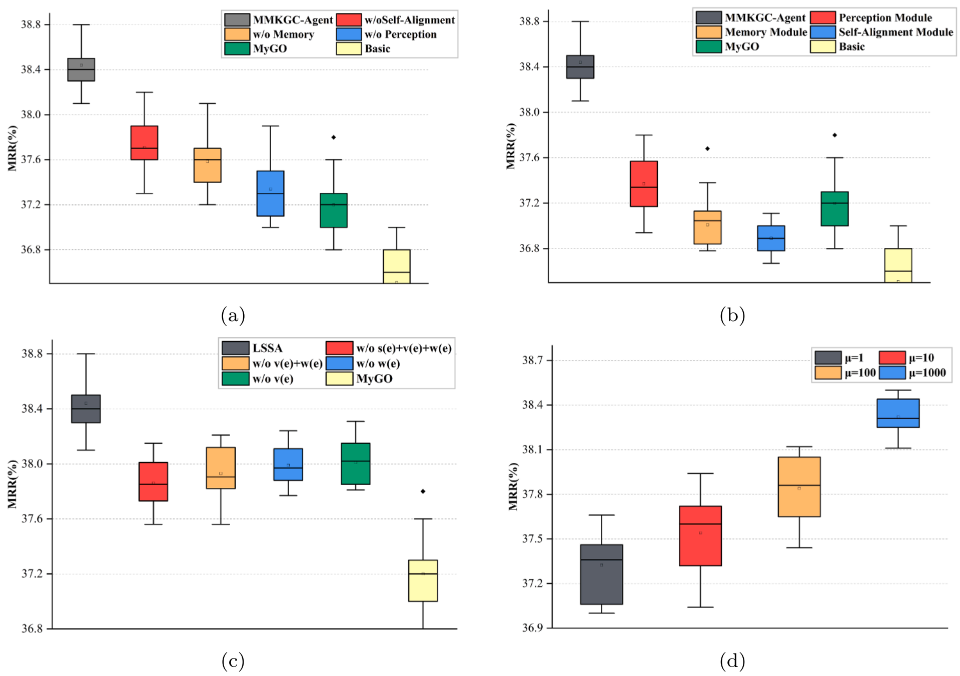


Fig. 3. The ablation study results on DB15K under different scenarios.

ership position, demonstrating that the unified design enhances overall robustness without sacrificing the ability to leverage basic structural cues.

In conclusion, the numerical improvements demonstrate that the closed-loop cognitive cycle is more than a modular collection of deep learning layers. By integrating perception, dynamic memory construction, and semantic stabilization into a mutually-reinforcing system, the agent systematically improves decision confidence and robustness under partial multimodal observations. This validates the proposed paradigm shift from static component stacking to an adaptive, integrated agentic framework.

5.3. Ablation study (RQ2)

To directly address whether the performance gain stems from the closed-loop architecture or merely from the choice of components, we compared the full MMKGC-Agent against a Sequential-Stack baseline. This baseline employs the identical underlying components, namely Transformer layers, GATv2, and the Barlow Twins objective, yet integrates them through a traditional linear feed-forward concatenation in lieu of our proposed closed-loop memory interaction.

As shown in Table 3, the Sequential-Stack baseline consistently achieves the lowest performance across all four datasets. For instance, on DB15K, the MRR of the sequential baseline is 1.80% lower than that of the full agentic framework. This disparity provides empirical evidence that simply stacking high-performance encoders is insufficient for complex multimodal reasoning. The superior performance of the Full Model proves that the functional coupling between the LRMP and the stabilization mechanism is what drives the agent's ability to synthesize precise, query-dependent representations.

Further ablation experiments involving the systematic removal of individual modules demonstrate that all components are indispensable for achieving peak performance. The results are summarized in Table 3 and Fig. 3. Across all four datasets, the elimination of any single module resulted in substantial performance degradation, corroborating the criticality of the synergistic Perception → Memory → Self-Alignment → Reasoning pipeline. These findings quantitatively substantiate the significance of the agent's closed-loop architectural design.

The second set of experiments investigates the impact of the self-alignment weight μ. Adjusting μ across $\{1, 10, 100, 1000\}$, we observed optimal performance at $\mu = 1000$ for DB15K and $\mu = 100$ for MKG-W. For the MKG-Y dataset, a smaller $\mu = 10$ was found to be optimal, reflecting its relatively sparse entity relations and limited multimodal information; a lower LSSA weighting suffices for semantic purification without over-regularizing global reasoning. In contrast, KVC16K, which contains dense entity relations and rich multimodal features, benefits from a high μ to fully exploit cross-modal semantic alignment and reinforce global reasoning. The difference in optimal μ values across datasets highlights the need to adjust the LSSA weight according to dataset characteristics to balance intra-modal purification and global reasoning effectively.

In summary, the ablation study provides strong evidence that the integration of perception, memory, self-alignment, and reasoning within a closed-loop cognitive framework is the key driver of MMKGC-Agent's superior performance, and that proper weighting of LSSA according to dataset characteristics is crucial for maximizing performance.

5.4. Exploration on perception module (RQ3)

To determine the optimal sensory mechanism for the agent's backbone, we conduct an in-depth analysis of its ability to resolve global–local structural signals. We posit that a cognitive agent requires not just passive reception of structural data, but an active attentional mechanism to discriminate between informative and noisy neighbors. To validate this, we substitute the local aggregation component of the Perception Module with five representative GNN architectures—GATv2, GATv1, GraphSAGE, GCN, and GIN—and evaluate their impact on the agent's reasoning behavior.

The results, visualized in Fig. 4, reveal a clear hierarchy of sensory acuity. Attention-based mechanisms (GAT variants) consistently outperform static aggregators (GCN, GraphSAGE, GIN). This indicates that treating all structural neighbors equally (or purely based on topology) is insufficient for multimodal reasoning; the agent must actively weigh the importance of neighbors based on their semantic relevance.

Crucially, GATv2 achieves the strongest performance. unlike standard GAT, which learns a static attention distribution, GATv2 employs a strictly dynamic attention mechanism that modifies connection weights

Table 3
Ablation study results on DB15K, MKG-W, MKG-Y, and KVC16K datasets. Both **Sequential-Stack** and **Full Model** are optimized using the **Barlow Twins** strategy with dataset-specific μ values ($\mu = 10$ for DB15K, $\mu = 100$ for MKG-W/MKG-Y, and $\mu = 1000$ for KVC16K). **Sequential-Stack** serves as a non-agentic baseline that employs identical components (Transformer, GAT, and Barlow Twins) but integrates them via simple linear concatenation instead of our proposed closed-loop memory interaction.

Setting		DB15K				MKG-W				MKG-Y				KVC16K			
		MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10
Arch. Synergy	**Perception Module**	37.37	29.77	41.13	52.17	36.54	29.81	39.01	48.24	37.44	35.81	38.23	42.99	14.55	7.43	14.33	25.98
	Memory Module	37.01	29.55	40.28	51.25	36.29	30.29	38.61	47.37	37.55	36.49	38.37	43.01	14.43	7.75	14.45	25.99
	Self-Alignment Module	36.89	28.99	40.68	52.01	36.44	30.29	38.69	47.94	37.41	36.25	38.11	42.98	14.54	7.35	14.32	25.94
	w/o Perception	37.28	29.70	40.68	51.94	35.81	29.73	37.89	47.13	37.56	36.08	38.33	43.23	14.75	7.77	14.55	26.01
	w/o Memory Module	37.59	29.56	41.56	52.82	36.78	30.77	39.07	48.25	37.78	35.77	38.08	43.26	14.65	7.62	14.37	25.97
	w/o Self-Alignment	37.66	29.95	41.50	52.24	36.63	30.10	39.48	48.40	37.95	36.10	38.48	43.34	14.76	7.84	14.97	26.10
Self- Alignment	**w/o** $s(e)+v(e)+w(e)$	37.86	29.93	41.75	52.86	36.37	29.87	38.88	48.05	38.14	35.76	38.72	43.76	14.87	7.98	15.15	26.68
	w/o $v(e)+w(e)$	37.93	29.97	41.87	52.90	36.47	29.99	38.96	48.11	38.55	36.25	38.99	43.91	15.32	8.76	15.75	27.08
	w/o $w(e)$	37.99	30.06	41.96	53.04	36.76	30.21	39.31	48.24	38.75	36.37	39.66	44.17	15.87	9.15	16.31	27.28
	w/o $v(e)$	38.01	30.14	42.04	53.08	36.78	30.27	39.41	48.33	38.99	36.42	39.54	44.31	15.98	9.27	16.41	27.33
	$\mu = 1$	37.36	29.75	41.38	52.29	36.17	29.76	38.68	47.86	39.05	36.66	41.12	44.75	16.17	9.76	16.68	27.86
	$\mu = 10$	37.54	29.90	41.17	52.45	36.58	30.48	39.14	48.43	**39.43**	**37.05**	**41.56**	**45.88**	16.47	9.91	17.14	28.43
	$\mu = 100$	37.84	30.27	41.33	52.52	**37.16**	**30.95**	**39.83**	**48.84**	39.05	36.37	41.23	45.57	16.53	10.01	17.62	28.47
	$\mu = 1000$	**38.32**	**30.38**	**42.44**	**53.25**	36.54	30.28	39.11	48.14	38.97	36.12	40.99	44.11	**16.67**	**10.34**	**18.85**	**29.52**
	InfoNCE	37.20	29.61	40.84	51.65	35.74	29.34	38.10	47.52	38.07	35.64	40.59	43.81	15.74	8.32	16.47	27.61
	Barlow Twins	**38.32**	**30.38**	**42.44**	**53.25**	**37.16**	**30.95**	**39.83**	**48.84**	**39.43**	**37.05**	**41.56**	**45.88**	**16.67**	**10.34**	**18.85**	**29.52**
Sequential-Stack		36.52	28.41	40.02	51.10	35.61	28.77	37.94	47.02	37.12	35.54	37.88	42.10	13.98	6.88	13.84	24.55
Full Model		**38.32**	**30.38**	**42.44**	**53.25**	**37.16**	**30.95**	**39.83**	**48.84**	**39.43**	**37.05**	**41.56**	**45.88**	**16.67**	**10.34**	**18.85**	**29.52**

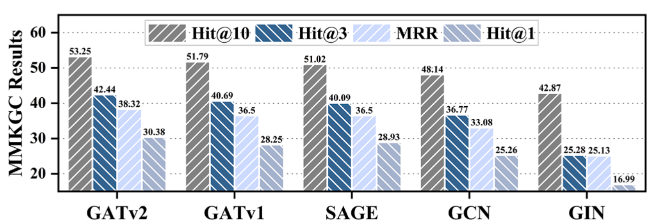


Fig. 4. Performance comparison of different neighborhood aggregation strategies within the perception module on DB15K.

based on the specific interaction between the central entity and its neighbors. This capability mirrors human-like selective attention, enabling the agent to filter out structural noise and refine local semantics while maintaining coherence with the global topology.

These findings confirm that the global–local dual-path design is a fundamental cognitive necessity. By dynamically adjusting attention across structural neighborhoods, the perception module ensures that the Memory Module receives high-fidelity, structurally refined "raw materials," which is the prerequisite for robust downstream memory construction and reasoning.

5.5. Exploration on memory module (RQ4)

To investigate the relationship between reasoning depth and decision precision, we conduct a parameter study on the maximum interaction depth (K) within the LRMP module. Rather than functioning as a simple residual operator, LRMP acts as a recursive refinement mechanism: it repeatedly performs interaction operations between the entity memory state and the relation, accumulating short-term structural patterns into an increasingly enriched cognitive state.

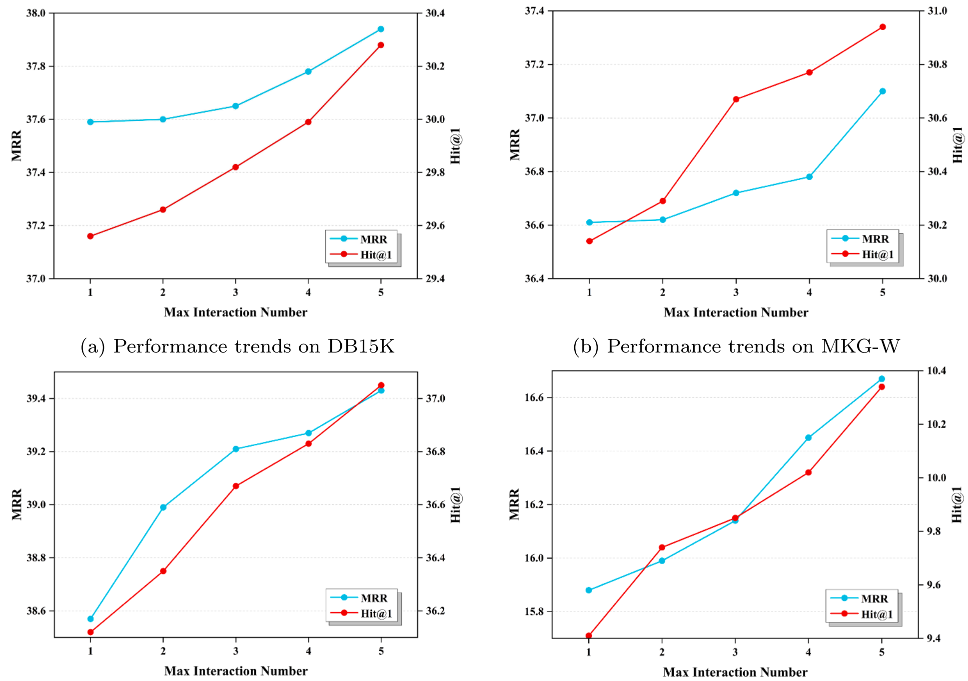
Fig. 5 visualizes the performance trajectory on the four datasets as the interaction depth increases from 1 to 5. while both exhibit an upward trend, the improvement in **Hits@1** is significantly sharper than that in MRR. Specifically, on the DB15K dataset, Hits@1 improves from approximately 29.5% to 30.3% (a relative gain of ∼2.7%), whereas the growth in MRR is more gradual. This disparity reveals the true nature of the memory mechanism: it contributes less to "general ranking" (finding the neighborhood) and more to "precision disambiguation" (identifying the exact target). With fewer interaction steps, the agent captures the broad structural context, which is sufficient to place the correct entity near the top (maintaining a decent MRR) but often fails to distinguish it from similar distractors. As the interaction cycles increase, the memory module iteratively filters out these plausible but incorrect candidates. This process effectively sharpens the decision boundary, allowing the agent to convert "top-ranking guesses" into "top-1 certainties." Therefore, the LRMP does not merely stabilize training; it equips the agent with the capacity for progressive disambiguation, ensuring that deeper deliberation leads to more precise outcomes.

5.6. Exploration on self-alignment module (RQ5)

To further dissect the stabilizing role of the LSSA module, we conducted fine-grained ablation studies on both the composition of the positive alignment set $C(e)$ and the choice of the underlying optimization objective. First, we evaluated the impact of different semantic views within the alignment set $C(e) = \{e_{\text{sec}}, s(e), v(e), w(e)\}$. As shown in Table 3, the comprehensive integration of global structural cues ($s(e)$), visual features ($v(e)$), and textual representations ($w(e)$) yields the highest performance. Removing any single view weakens the agent's internal consistency. This demonstrates that LSSA functions as a holistic semantic anchor: by forcing these heterogeneous signals to align, the module neutralizes modality-specific noise, ensuring that the agent's memory is built upon a robust, multifaceted understanding of the entity rather than biased, single-source features.

We further compared the proposed redundancy-reduction objective (Barlow Twins) against the traditional contrastive objective (InfoNCE). The results in Table 3 reveal that Barlow Twins achieves consistently superior performance. We attribute this to the fundamental difference in their mechanisms: InfoNCE relies on negative sampling to perform instance discrimination, which carries the risk of "false negatives" (repelling semantically similar entities) and depends heavily on batch size. Barlow Twins, conversely, focuses on feature disentanglement. By minimizing the off-diagonal terms of the cross-correlation matrix, it forces the learned feature dimensions to be orthogonal (non-redundant).

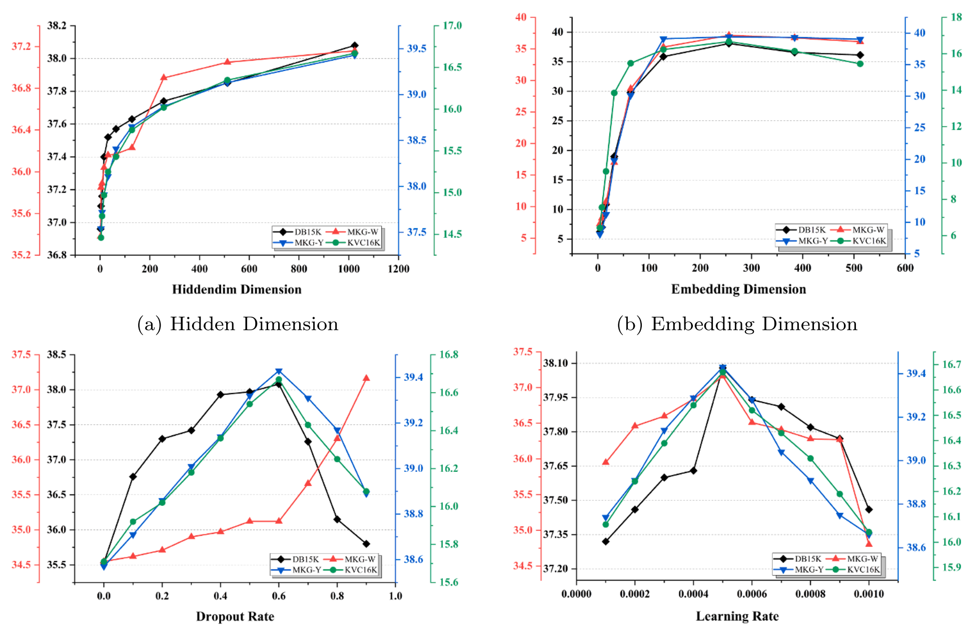
This property is critical for the agent's cognitive loop: it provides the subsequent Memory Module with clean, disentangled feature axes, allowing the LRMP to manipulate distinct semantic attributes without interference, thereby ensuring stable and precise reasoning.



(a) Performance trends on DB15K (b) Performance trends on MKG-W

(c) Performance trends on MKG-Y (d) Performance trends on KVC16K

Fig. 5. Interaction depth analysis of the Learnable Residual Memory Pathway across four heterogeneous datasets.



(a) Hidden Dimension (b) Embedding Dimension

(c) Dropout Rate (d) Learning Rate

Fig. 6. Sensitivity analysis of MMKGC-Agent with respect to hidden dimension, embedding dimension, dropout rate, and learning rate, evaluated by MRR on DB15K and MKG-W. The x-axis denotes different hyperparameters, and the y-axis shows MRR.

5.7. Sensitivity analysis for hyperparameter choices (RQ6)

To assess the adaptability of the proposed MMKGC-Agent, we further conduct a sensitivity analysis with respect to four key hyper-parameters: hidden dimension, embedding dimension, dropout rate, and learning rate. This analysis reflects how the agent's cognitive loop responds to different capacity and regularization settings, thereby offering insight into its robustness under varying environmental conditions. Representative results are shown in Fig. 6.

Hidden Dimension. The experimental results indicate that the Mean Reciprocal Rank (MRR) steadily increases with the hidden dimension, peaking at 1024. At smaller values (e.g., 4, 8, 16), the model's expressive capacity is severely constrained, leading to low MRR. Unlike the embedding dimension, enlarging the hidden dimension does not significantly harm performance, suggesting that the agent effectively leverages the additional capacity without accumulating excessive noise.

Embedding Dimension. As shown in Fig. 6, the MRR improves as the embedding dimension increases, with the best performance at 256.

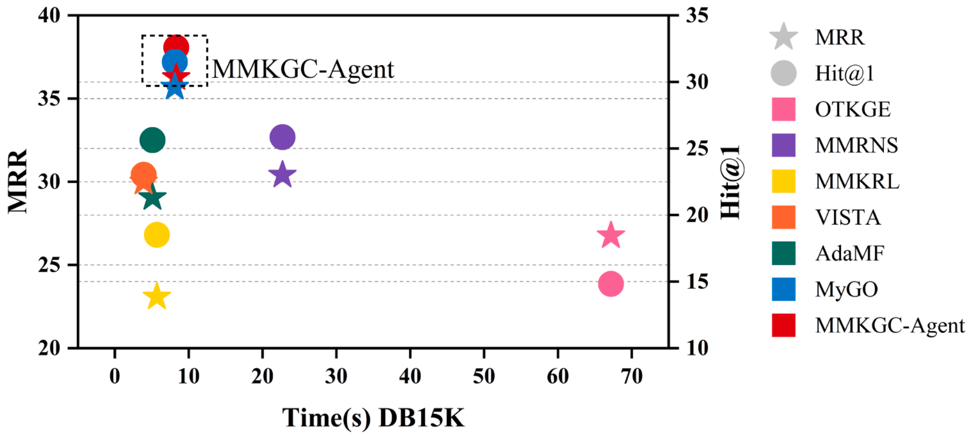


Fig. 7. The results of the efficiency experiment. We report the MRR and Hit@1 results on the DB15K datasets.

Further enlargement to 384 and 512 results in performance degradation, indicating that overly high embedding dimensionality introduces redundancy and weakens semantic purity. This demonstrates that a moderate embedding size is optimal for balancing expressiveness and semantic alignment.

Dropout Rate. The dropout rate plays a crucial role in regulating the agent's ability to generalize across multimodal cues. On DB15K, the peak MRR (38.32) occurs at a dropout rate of 0.6, while on MKG-W, MKG-Y andKVC16K, the optimal value is reached at 0.9. Moderate dropout effectively prevents over-fitting, but excessively high dropout hampers representation learning, confirming that regularization must be carefully tuned to maintain cognitive stability.

Learning Rate. The learning rate strongly influences convergence and stability. On both datasets, MRR improves as the rate increases, reaching an optimal point around 5×10^{-4}. Beyond this threshold, performance declines, and at 0.001 the model becomes unstable due to overly aggressive gradient updates. These results suggest that a learning rate near 5×10^{-4} provides a favorable balance between convergence efficiency and reasoning accuracy.

Overall, the sensitivity analysis shows that the MMKGC-Agent maintains stable performance across a range of hyper-parameter choices, while achieving peak results under specific configurations. This robustness underscores its adaptability and indicates that its cognitive framework is not overly dependent on fragile parameter settings.

5.8. *Efficiency analysis (RQ7)*

To evaluate the computational efficiency of MMKGC-Agent in conjunction with its cognitive effectiveness, we conducted dedicated runtime measurements. As shown in Fig. 7, which reports results on the DB15K dataset, MMKGC-Agent achieves strong MRR and $Hit@1$ performance while maintaining a competitive per-epoch training time. On DB15K, each training epoch requires approximately 8.1 s, which is comparable to that of the fastest baseline method, VISTA, while yielding notably improved predictive accuracy. These results indicate that the proposed closed-loop cognitive architecture does not incur additional computational overhead and is suitable for practical MMKGC applications.

In addition to DB15K, we further measured the training efficiency of MMKGC-Agent on three additional benchmarks. On the medium-scale MKG-W and MKG-Y datasets, the per-epoch training times are 6.0 s and 3.8 s, respectively. To assess scalability on larger graphs, we also evaluated MMKGC-Agent on the large-scale KVC16K dataset, where a single training epoch requires approximately 24 s. Although the results on these datasets are not visualized in Fig. 7, the measured runtimes consistently demonstrate stable scaling behavior as the graph size increases.

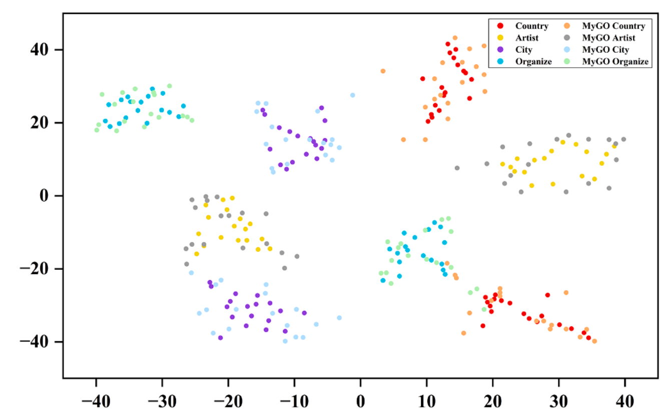


Fig. 8. Embedding visualization results.

To provide theoretical insight into the observed scalability, we analyze the computational complexity of MMKGC-Agent. Let $|T|$ denote the number of triples in the multimodal knowledge graph, d the embedding dimension, and $\bar{k}$ the average one-hop neighborhood size. The global multimodal perception encoder processes bounded-length text–image sequences for each triple with a computational cost of $O(|T| \cdot d)$. The local graph encoder applies dynamic attention within one-hop neighborhoods, resulting in a complexity of $O(|T| \cdot \bar{k} \cdot d)$. The residual memory pathway and the self-alignment module consist of fixed-depth transformations and redundancy reduction operations, which together contribute an additional cost of $O(|T| \cdot d)$. Finally, the TuckER-style reasoning module scores each triple through bilinear interactions, yielding an overall decoding complexity of $O(|T| \cdot d^2)$. Since both d and $\bar{k}$ remain bounded in practical MMKG settings, the total per-epoch training time scales linearly with $|T|$.

Overall, the empirical results on DB15K, MKG-Y, MKG-W, and KVC16K, together with the complexity analysis, indicate that MMKGC-Agent achieves an effective balance between enhanced multimodal reasoning capability and practical computational scalability.

5.9. *Case study (RQ8)*

To further illustrate the effectiveness of the proposed MMKGC-Agent, we conduct a case study on selected entities from DB15K. Four representative categories—artist, city, organization, and country—are chosen, with two entities sampled per category. For each entity, the multimodal sequence e' is tokenized into semantic atoms, and t-SNE is applied to project their embeddings into a 2D space (Fig. 8). The resulting visualization shows that MMKGC-Agent produces compact entity-specific

Table 4
Query-Adaptive Scores of MMKGC-Agent vs. Static MyGO.

Method	Head	Relation	Tail	S(h,r,t)	Cosine sim. w/h
MyGO	$\tilde{h}$(Amazon)	*headquarters*	$\tilde{t}$(Seattle)	0.68	-
Ours	$\tilde{h}''$(Amazon)	*headquarters*	$\tilde{t}''$(Seattle)	0.97	-
Ours	$\tilde{h}''$(Apple)	*headquarters*	$\tilde{t}''$(Cupertino)	-	0.85
Ours	$\tilde{h}''$(Apple)	*product category*	$\tilde{t}''$(Electronics)	-	0.72

clusters with clear inter-entity separation. Even entities within the same category exhibit distinct internal structures, demonstrating the model's ability to capture both fine-grained local interactions and global semantic context.

To highlight the advantage over baseline methods, we visualize MyGO under identical t-SNE settings. As shown in Fig. 8, MyGO generates mixed and overlapping clusters—especially among visually similar countries and organizations—suggesting limited capacity for isolating entity-specific semantics. In contrast, MMKGC-Agent achieves significantly stronger intra-class cohesion and inter-class separation, confirming that the closed-loop Perception → Memory → Self-Alignment → Reasoning pipeline yields more discriminative multimodal representations.

We further examine ambiguous multimodal triples such as (*Amazon*, *headquarters*, *Seattle*). Baselines like MyGO, lacking local memory of fine-grained evidence, are easily influenced by irrelevant visual cues and thus produce low-confidence predictions (score 0.68). MMKGC-Agent, however, retrieves discriminative local signals—such as company logos and product descriptions—and the LSSA module reinforces their relevance while suppressing noise, resulting in a purified embedding e' and a correct prediction with high confidence (score 0.97). This trend aligns with Table 4, where MMKGC-Agent consistently yields higher triple scores on ambiguous queries.

Finally, we analyze MMKGC-Agent's dynamic reasoning capability by inspecting query-dependent memory states $\tilde{\mathbf{h}}''$. For the entity "Apple," the agent generates two distinct memory representations for the relations r_1 ("*headquarters*") and r_2 ("*product category*"). Unlike static KGE models, which use a fixed embedding, MMKGC-Agent produces two clearly differentiated memory states with a high Euclidean distance of 0.51.Furthermore, we note that the cosine similarity between the original entity embedding $\mathbf{h}$ and the two generated memory states is 0.85 and 0.72 for r_1 and r_2, respectively, underscoring the preservation of global identity while achieving context-specific adaptation. These states bias the representation toward geographic cues for r_1 and product features for r_2, enabling accurate predictions of *Cupertino* and *CElectronics*, respectively. This confirms that the agent performs genuine query-dependent semantic adaptation.

Overall, this case study shows that MMKGC-Agent's superiority stems not only from enhanced multimodal perception but also from its local-memory-driven semantic refinement. The synergy among perception, memory, self-alignment, and reasoning yields robust, fine-grained, and context-sensitive embeddings, enabling superior knowledge completion performance in complex multimodal environments.

6. Conclusion

In this paper, we introduced the MMKGC-Agent, a framework that fundamentally shifts the paradigm of Multimodal Knowledge Graph Completion from static representation matching to dynamic cognitive reasoning. By establishing a unified closed-loop cognitive architecture that integrates perception, memory, and stabilization, the agent successfully overcomes the inherent rigidity of conventional embedding models. The core innovation lies in the functional interdependency between the Learnable Residual Memory Pathway and the Self-Alignment mechanism, which enables the agent to actively synthesize query-dependent memory states while maintaining global semantic coherence. This synergistic design represents a non-trivial advancement over simple sequential stacking by effectively balancing representational plasticity with intrinsic stability.

Empirical evaluations across four diverse benchmarks validate the superiority of this approach, demonstrating remarkable robustness against multimodal noise and exceptional precision in resolving semantic ambiguities. Notably, our complexity analysis reveals that despite its cognitive richness, the agent maintains a favorable computational efficiency with a linear complexity of $O(|T|d^2)$, striking an optimal balance between reasoning depth and scalability. While we acknowledge certain limitations, such as the sensitivity to the stabilization weight μ and the memory footprint when processing ultra-large-scale graphs, these constraints are counterbalanced by the agent's superior predictive accuracy and adaptive capabilities. Future research will focus on developing automated hyperparameter optimization strategies and enhancing the scalability of the agentic reasoning process for massive knowledge environments, ultimately extending this paradigm to complex autonomous reasoning tasks.

CRediT authorship contribution statement

Kai Yang: Conceptualization, Methodology, Software, Formal analysis, Experimental investigation, Visualization, Writing – original draft, Writing – review & editing; **Xinzhi Wang**: Data curation, Software, Validation, Writing – review & editing; **Baoguo Lu**: Data curation, Validation, Writing – review & editing; **Xiangfeng Luo**: Supervision, Methodological guidance, Writing – review & editing, Funding acquisition; **Chengfan Li**: Supervision, Resources, Project administration, Funding acquisition; **Jianqiang Huang**: Investigation, Results analysis, Writing – review & editing.

Data avalibility

I have shared the link to the repository. All datasets and source code are openly available at https://github.com/KaiYang2001/MMKGC-Agent.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research reported in this paper was supported in part by the National Nature Science Foundation of China under the grant No.61991415.

References

Bo, X., Zhang, Z., Dai, Q., Feng, X., Wang, L., Li, R., Chen, X., & Wen, J.-R. (2024). Reflective multi-agent collaboration based on large language models. In *Advances in neural information processing systems (neurIPS) 2024*. NeurIPS 2024 poster / OpenReview preprint https://openreview.net/pdf?id=wWiAR5mqXq.

Bordes, A., Usunier, N., Garcia-Durán, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. In *Proceedings of the 27th international conference on neural information processing systems - volume 2* NIPS'13 (p. 2787–2795). Red Hook, NY, USA: Curran Associates Inc.

Cao, X., Shi, Y., Wang, J., Yu, H., Wang, X., & Yan, Z. (2022a). Cross-modal knowledge graph contrastive learning for machine learning method recommendation. In *Proceedings of the 30th ACM international conference on multimedia* MM '22 (p. 3694–3702). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3503161.3548273

Cao, Z., Xu, Q., Yang, Z., He, Y., Cao, X., & Huang, Q. (2022b). Otkge: multi-modal knowledge graph embeddings via optimal transport. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh (Eds.), *Advances in neural information processing systems* (pp. 39090–39102). Curran Associates, Inc. (*vol. 35*). https://proceedings.neurips.cc/paper_files/paper/2022/file/ffdb280e7c7b4c4af30e04daf5a84b98-Paper-Conference.pdf.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., & Joulin, A. (2020). Unsupervised learning of visual features by contrasting cluster assignments. In *Proceedings of the 34th international conference on neural information processing systems* NIPS '20. Red Hook, NY, USA: Curran Associates Inc.

Chen, L., Li, Z., Wang, Y., Xu, T., Wang, Z., & Chen, E. (2020a). Mmea: entity alignment for multi-modal knowledge graph. In *International conference on knowledge science, engineering and management* (pp. 134–147). Springer.

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020b). A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th international conference on machine learning* ICML'20. JMLR.org.

Chen, Z., Zhang, W., Huang, Y., Chen, M., Geng, Y., Yu, H., Bi, Z., Zhang, Y., Yao, Z., Song, W., Wu, X., Yang, Y., Chen, M., Lian, Z., Li, Y., Cheng, L., & Chen, H. (2023). Tele-knowledge pre-training for fault analysis. In *ICDE* (pp. 3453–3466). IEEE.

Chen, Z., Zhang, Y., Fang, Y., Geng, Y., Guo, L., Chen, X., Li, Q., Zhang, W., Chen, J., Zhu, Y., Li, J., Liu, X., Pan, J. Z., Zhang, N., & Chen, H.-z. (2024). Knowledge graphs meet multi-modal learning: A comprehensive survey. https://api.semanticscholar.org/CorpusID:267547866, arXiv:2402.05391.

Cheng, P., Hao, W., Dai, S., Liu, J., Gan, Z., & Carin, L. (2020). Club: a contrastive log-ratio upper bound of mutual information. In *Proceedings of the 37th international conference on machine learning* ICML'20. JMLR.org.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K., et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 conference of the north American chapter of the association for computational linguistics: Human language technologies, volume 1 (long and short papers)* (pp. 4171–4186). Minneapolis, Minnesota: Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Gong, B., Tan, S., Feng, Y., Xie, X., Li, Y., Chen, C., Zheng, K., Shen, Y., & Zhao, D. (2024). Uknow: a unified knowledge protocol with multimodal knowledge graph datasets for reasoning and vision-language pre-training. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, & C. Zhang (Eds.), *Advances in neural information processing systems* (pp. 9612–9633). Curran Associates, Inc. (*vol. 37*). https://proceedings.neurips.cc/paper_files/paper/2024/file/126784e4d5a92afff92d13aee155554b-Paper-Datasets_and_Benchmarks_Track.pdf.

Guo, S., Deng, C., Wen, Y., Chen, H., Chang, Y., & Wang, J. (2024). DS-Agent: automated data science by empowering large language models with case-based reasoning. In *Forty-first international conference on machine learning*. https://openreview.net/forum?id=LfJgeBNCFI.

He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning. In *2020 IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 9726–9735). https://doi.org/10.1109/CVPR42600.2020.00975

Hong, S., Liu, Y., Li, Z., Li, S., & He, Y. (2024). Multi-agent collaborative perception via motion-aware robust communication network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 15301–15310).

Jian, Y., Luo, X., Li, Z., Zhang, M., Zhang, Y., Xiao, K., & Hou, X. (2025a). Apkgc: noise-enhanced multi-modal knowledge graph completion with attention penalty. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(14), 15005–15013. https://ojs.aaai.org/index.php/AAAI/article/view/33645. https://doi.org/10.1609/aaai.v39i14.33645

Jian, Y., Zhang, M., Qin, Z., Xie, C., Xiao, K., Zhang, Y., & Li, Z. (2025b). Adaptive modality interaction transformer for multimodal knowledge graph completion. *ACM Transactions on Knowledge Discovery From Data*, *19*(8). https://doi.org/10.1145/3760786

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. In *Proceedings of the 34th international conference on neural information processing systems* NIPS '20. Red Hook, NY, USA: Curran Associates Inc.

Lee, J., Chung, C., Lee, H., Jo, S., & Whang, J. (2023). VISTA: Visual-textual knowledge graph representation learning. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Findings of the association for computational linguistics: EMNLP 2023* (pp. 7314–7328). Singapore: Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.findings-emnlp.488

Li, X., Zhao, X., Xu, J., Zhang, Y., & Xing, C. (2023). Imf: interactive multimodal fusion model for link prediction. In *Proceedings of the ACM web conference 2023* WWW '23 (p. 2572–2580). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3543507.3583554

Li, Z., Huang, W., Gong, X., Luo, X., Xiao, K., Deng, H., Zhang, M., & Zhang, Y. (2025). Decoupled semantic graph neural network for knowledge graph embedding. *Neurocomputing*, *611*, 128614. https://www.sciencedirect.com/science/article/pii/S0925231224013857. https://doi.org/10.1016/j.neucom.2024.128614

Liu, Y., Li, H., Garcia-Duran, A., Niepert, M., Onoro-Rubio, D., & Rosenblum, D. S. (2019). Mmkg: multi-modal knowledge graphs. In *The semantic web: 16th international conference, ESWC 2019, portorož, slovenia, june 2–6, 2019, proceedings* (p. 459–474). Berlin, Heidelberg: Springer-Verlag. https://doi.org/10.1007/978-3-030-21348-0_30

Mousselly-Sergieh, H., Botschen, T., Gurevych, I., & Roth, S. (2018a). A multimodal translation-based approach for knowledge graph representation learning. In M. Nissim, J. Berant, & A. Lenci (Eds.), *Proceedings of the seventh joint conference on lexical and computational semantics* (pp. 225–234). New Orleans, Louisiana: Association for Computational Linguistics. https://doi.org/10.18653/v1/S18-2027

Mousselly-Sergieh, H., Botschen, T., Gurevych, I., & Roth, S. (2018b). A multimodal translation-based approach for knowledge graph representation learning. In M. Nissim, J. Berant, & A. Lenci (Eds.), *Proceedings of the seventh joint conference on lexical and computational semantics* (pp. 225–234). New Orleans, Louisiana: Association for Computational Linguistics. https://doi.org/10.18653/v1/S18-2027

Sun, R., Cao, X., Zhao, Y., Wan, J., Zhou, K., Zhang, F., Wang, Z., & Zheng, K. (2020). Multi-modal knowledge graphs for recommender systems. In *Proceedings of the 29th ACM international conference on information & knowledge management* CIKM '20 (p. 1405–1414). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3340531.3411947

Trirat, P., Jeong, W., & Hwang, S. J. (2025). Automl-agent: a multi-agent LLM framework for full-pipeline automl. In *Proceedings of the 2025 international conference on machine learning (ICML 2025)*. https://arxiv.org/abs/2410.02958.

Wang, B., Zhang, L., Wang, Z., Zhao, Y., & Zhou, T. (2023a). Core: cooperative reconstruction for multi-agent perception. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 8710–8720).

Wang, M., Wang, S., Yang, H., Zhang, Z., Chen, X., & Qi, G. (2021). Is visual context really helpful for knowledge graph? a representation learning perspective. In *Proceedings of the 29th ACM international conference on multimedia* MM '21 (p. 2735–2743). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3474085.3475470

Wang, X., Meng, B., Chen, H., Meng, Y., Lv, K., & Zhu, W. (2023b). Tiva-kg: a multimodal knowledge graph with text, image, video and audio. In *Proceedings of the 31st ACM international conference on multimedia* MM '23 (p. 2391–2399). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3581783.3612266

Wang, Z., Li, L., Li, Q., & Zeng, D. (2019). Multimodal data enhanced representation learning for knowledge graphs. In *2019 international joint conference on neural networks (IJCNN)* (pp. 1–8). https://doi.org/10.1109/IJCNN.2019.8852079

Xie, R., Liu, Z., Luan, H., & Sun, M. (2017). Image-embodied knowledge representation learning. In *Proceedings of the twenty-sixth international joint conference on artificial intelligence, IJCAI-17* (pp. 3140–3146). https://doi.org/10.24963/ijcai.2017/438

Xu, D., Xu, T., Wu, S., Zhou, J., & Chen, E. (2022). Relation-enhanced negative sampling for multimodal knowledge graph completion. In *Proceedings of the 30th ACM international conference on multimedia* MM '22 (p. 3857–3866). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3503161.3548388

Zhang, Y., Chen, M., & Zhang, W. (2023). Modality-aware negative sampling for multi-modal knowledge graph embedding. In *2023 international joint conference on neural networks (IJCNN)* (pp. 1–8). https://doi.org/10.1109/IJCNN54540.2023.10191314

Zhang, Y., Chen, Z., Guo, L., Xu, Y., Hu, B., Liu, Z., Zhang, W., & Chen, H. (2024a). Native: Multi-modal knowledge graph completion in the wild. https://arxiv.org/abs/2406.17605.

Zhang, Y., Chen, Z., Guo, L., Xu, Y., Hu, B., Liu, Z., Zhang, W., & Chen, H. (2025). Tokenization, fusion, and augmentation: towards fine-grained multi-modal entity representation. In *AAAI* (pp. 13322–13330). AAAI Press.

Zhang, Y., Chen, Z., Liang, L., Chen, H., & Zhang, W. (2024b). Unleashing the power of imbalanced modality information for multi-modal knowledge graph completion. https://arxiv.org/abs/2402.15444.